



Alpha emitter isotopes and PSMA ligands: the near future therapeutic prospective for castration-resistant prostate cancer

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Published online: 23 December 2023

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The theragnostic approach was proposed with the introduction of 131-Iodine for the treatment of both benign and malign thyroid diseases [1–5]. The therapeutic effect of 131-Iodine is based on its ability to emit β -particles, which are characterized by a high linear energy transfer (LET) and a relatively long range of penetration. The β -emission increases the average dose to the tumor, but it also results in irradiation of surrounding healthy tissue [1–6]. During the last years, [¹⁷⁷Lu]Lu-PSMA-617 has been introduced as a promising and effective therapy in patients with prostate cancer and metastatic disease [7–9]. Similarly, to 131-Iodine, ¹⁷⁷Lu is a typical theragnostic agent, able to emit both β -particles and γ photons. The use of [¹⁷⁷Lu]Lu-PSMA-617 has been previously tested in several populations with metastatic castration-resistant prostate cancer (mCRPC). VISION study [7], an important multicenter phase 3 randomized trial of 831 mCRPC patients, demonstrated that [¹⁷⁷Lu]Lu-PSMA-617 plus standard of care (SOC) significantly prolonged radiological progression-free survival (rPFS) (8.7 vs. 3.4, months) and overall survival (OS) (15.3 vs. 11.3, months) compared to control group of SOC patients. In addition, the quality of life was not adversely affected by this innovative and revolutionary therapy.

However, it should be considered that patients with mCRPC disease can be heterogeneous, and bone metastases can be observed in almost all patients [10–12]. Therefore, the availability of therapeutic agents able to preserve the healthy tissues is particularly important in these patients. Targeted alpha therapy (TAT) has been introduced to overcome those limitations. The α -emitter radioisotopes have optimal physical characteristics, such as high linear energy transfer (LET) and a short penetration range. Thus,

223-Radium has already been introduced for the palliative treatment of bone metastases in patients with mCRPC, demonstrating significant benefit on symptoms, but not on the OS in these patients [10–12].

More recently, 225-Actinium has demonstrated an encouraging therapeutic option in patients with mCRPC, thanks to its optimal half-life (9.9 days) and short penetration ranges (400–100 μ m) [13–15].

The recent introduction in clinical use of [²²⁵Ac] Ac-PSMA-617, [²²⁵Ac] Ac-DOTATOC, [²²⁵Ac] Ac-DOTA-substance-P reported significantly improved responses in patients with prostate cancer, neuroendocrine tumors, and gliomas, respectively [16–18].

According to that evidence, we read with great interest a recent study by Ballal and coworkers [19], exploring the outcome of 56 patients with end-stage metastatic mCRPC treated with [²²⁵Ac] Ac-PSMA-617. Overall survival and rPFS, biochemical response through prostate specific antigen (PSA) blood levels, and the toxicity were considered endpoints.

The enrolled patients were stratified into two groups according to previous treatments: 27 patients were refractory to [¹⁷⁷Lu]Lu-PSMA-617 and 29 were [¹⁷⁷Lu]Lu-PSMA-617 treatment-naïve. [²²⁵Ac]Ac-PSMA-617 was intravenously administered to all patients at intervals of 8 weeks between each cycle. The administered dose ranged from 100–150 KBq/kg body weight until a cumulative dose of 74 MBq for a minimum of 2 cycles of [²²⁵Ac]Ac-PSMA-617 therapy.

Ballal et al. [19] in their investigation found that the OS for the [¹⁷⁷Lu]Lu-PSMA-617 naïve patients was 15 months compared to 10 months for the [¹⁷⁷Lu]Lu-PSMA-617 refractory group. Moreover, the rPFS was 10 and 15 months for the refractory and naïve patients, respectively. Therefore, they found that both median OS and rPFS were improved using [²²⁵Ac]Ac-PSMA-617 TAT in pretreated mCRPC patients. Moreover, reasonable disease control and survival benefits were also observed even in patients [¹⁷⁷Lu]Lu-PSMA-617 refractory.

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According to biochemical responses, 56 patients showed a PSA decline of 50% or more. The PSA response was similar between the [^{177}Lu]Lu-PSMA-617-naïve group and the [^{177}Lu]Lu-PSMA-617-refractory group. The toxicity profile was also investigated, and the occurrence of fatigue was observed in 70% of the entire cohort, followed by loss of appetite (43%), myalgia (38%), and xerostomia (32%). However, these effects resolved within 5 weeks post-therapy. As already reiterated, the [^{225}Ac] Ac-PSMA-617 is a α -emitter radioisotope, thus it does not offer the possibility to perform a single photon-computed tomography/computed tomography (SPECT/CT) post-therapy scan. This aspect makes the use of positron emission tomography/computed tomography (PET/CT) necessary for the evaluation of these patients. In fact, they underwent a [^{68}Ga]Ga-PSMA PET/CT every 2–3 cycles to evaluate the staging of disease and treatment response [20–22].

The [^{225}Ac] Ac-PSMA-617 is a promising opportunity for patients with mCRPC. In particular, the adoption of this approach as the first line of therapy in patients who are candidates for radioligand therapy could be a great opportunity to obtain a significant impact on outcome without affecting healthy tissues. For this purpose, further studies are needed to standardize treatment protocols and improve the outcome of mCRPC patients.

Declarations

Ethical approval Institutional Review Board approval was not required because the paper is an Editorial.

Informed consent Not applicable.

Conflict of interest The authors declare no competing interests.

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